

XINYU CHEN

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EDUCATION BACKGROUND

FORDHAM UNIVERSITY, GABELLI SCHOOL OF BUSINESS

New York, NY

MS, Quantitative Finance

2022 - 2024

- Relevant Courses: Derivatives, Fixed Income, Machine Learning, Risk Management, Simulation Application, Financial Math

UNIVERSITY OF CALIFORNIA, SANTA BARBARA

Santa Barbara, CA

Bachelor of Arts, major in Economics, Bachelor of Arts, minor in Statistics and Data Science

2020 - 2022

- Relevant Courses: Macroeconomics, Microeconomics, Corporate Finance, Stochastic Calculus, Time Series, Regression

TECHNICAL SKILLS

- Programming & Analytical Tools: Python (data analysis, modular libraries), MySQL, EXCEL, VBA, Tableau
- Quantitative Skills: Monte Carlo simulation, Value-at-Risk, Model Validation, Yield Curve Construction, Workflow Automation, Derivative Pricing, Sensitivity Analysis

PROFESSIONAL EXPERIENCE

HYDE PARK INVESTMENT SERVICE

Remote

Market Risk Analyst Intern

2024.01 - 2024.09

- Developed a Python-based risk processing pipeline integrating **MSCI API** to automate **VaR, sensitivity analysis**, and **stress testing** workflows for equity portfolios
- Monitored rolling beta, correlation breakdowns, and factor tilt shifts to identify crowding, hidden beta, and **emerging style risk**
- Designed a **MySQL** database for efficient risk data storage and aggregation, and used **VBA** for risk reporting

GLOBAL AI

Manhattan, NY

Model Validation Analyst

2023.09 - 2023.12

- Validated FX and **equity exotic** pricing models, including barrier options and quanto digital options
- Conducted model verification using provided market data inputs and existing model frameworks, tested pricing consistency under volatility smile and yield curve assumptions
- Reviewed and debugged pricing model code to verify alignment with theoretical logic and ensure numerical accuracy
- Authored formal documentation outlining validation procedures, test results, and discrepancies for the model risk management team

ACADEMIC EXPERIENCE

Market Risk Analytics Library

2025.03 - Present

- Developed an object-oriented Python framework for multi-method market risk modeling, implementing **Historical VaR, Monte Carlo VaR, CVaR, and SVaR** with PCA + Copula-driven stress simulation
- Integrated Barra-style **factor exposure modeling**, shrinkage covariance estimation, and factor-specific risk decomposition to simulate style/sector attribution and crowding risk
- **Optimized code performance** and modularity by leveraging best practices in Python OOP, enhancing the system's scalability and maintainability

Fixed Income Analytics Library

2025.02 - 2025.05

- Architected a modular **object-oriented fixed income library** in Python, implementing valuation and risk metrics for bonds and swaps
- Developed advanced **yield curve construction** algorithms for Treasury and SOFR markets, featuring specialized **bootstrapping** techniques, Nelson-Siegel smoothing, and handling of on-the-run premiums
- Built sophisticated risk management capabilities including **duration/convexity metrics, DV01/PV01** sensitivities, and scenario analysis with proper handling of market conventions